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Declaration

Acknowledgements

Abstract

Chapter 1

Gravitational Wave Interferometers

1.1 Introduction

Gravitational wave astronomy offers a new, unique way to observe the universe. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. It is of great importance to astronomy that large numbers of these signals are measured, so it is imperative that gravitational wave detectors are made more sensitive.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferometer, with a particular emphasis on LIGO, can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the

sensitivity required to detect gravitational waves and so this is the focus of this chapter.

1.2 Astrophysical Origins of Gravitational Waves

Gravitational waves emanating BBH (e.g. [1,2]), BNS [3], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [4] have been detected; this is summarised in [5]. The waveform of gravitational radiation emitted from these systems is known as a chirp signal. These signals increase in frequency as the two objects get closer to merging. Due to the geometry of these systems, the frequency of the gravitational waves are in the audioband with the cut-off frequency, determined by the radius of the objects that will merge, between 100 Hz-10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein's general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo gravitational wave detectors [6]. Gravitational waves are emitted from mass quadrupole systems (see e.g. [7]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses involved.

Other sources of gravitational waves include pulsars [8, 9] and magnetars [10]. Asymmetries in these systems will cause them to emit a gravitational wave with a constant frequency. Gravitational waves will be emitted by supernovae [11]. Weaker sources that the future gravitational wave interferometers may detect in-

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 04/30/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	date?	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 1.1: table of observation runs

clude: the stochastic background of gravitational waves [12], similar to the the cosmic microwave background radiation, as well as a background level of gravitational waves due to the merging of compact binaries [13]; dark matter [14, 15] and cosmic strings [16].

To improve statistical models of the systems that emit gravitational waves, the number detections needs to be increased. To detect continuous wave signals, the signal to noise ratio improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, the gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades that increase the detector sensitivity to gravitational waves. The current observation runs are summed up in Table 1.1.

1.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [7]). The effect these waves have, in the local Lorentz frame, on test masses on Earth is to move them by a small amount ¹.

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These are referred to as *end test masses* (ETMs). A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 1.1. For test masses L away from the origin of the local Lorentz frame, the distance they move ΔL due to an incoming plus polarised gravitational wave is

$$\Delta L = Lh. \tag{1.1}$$

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: all physical laws that are expressible as a tensor equation in special relativity have the exact same form in a local Lorentz frame of a curved space time. In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is sufficiently small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

The interferometer is sensitive to changes in the phase between the two arms corresponding of the order of one wavelength,

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} Lh, \quad (1.2)$$

allowing for small distances when laser light is used. The choice of $\lambda_{\text{EM}} = 1064 \text{ nm}$ as the wavelength which the detector operates with is determined by the low loss and low thermal noise coatings [17] and test masses that can be fabricated for 1064 nm light.

The gravitational wave signal is encoded in intensity fluctuations of the light at the *antisymmetric* port (see Figure 1.1). Gravitational wave detectors are operated on the *dark fringe* meaning that there is a low amount of power at the antisymmetric port for the signal light to beat with and the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , is the differential arm length. At the beam splitter, the electric field, Ψ_{AS} , emerging from the antisymmetric port due to the interference of the beams, with differential phase $\phi = \phi_0 + \Delta\phi$, returning from the end test masses, Ψ_y and Ψ_x , is

$$\Psi_{\text{AS}} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (1.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i(\phi_0 + \Delta\phi)}}{\sqrt{2}} - \frac{e^{-i(\phi_0 + \Delta\phi)}}{\sqrt{2}} \right). \quad (1.4)$$

The intensity of the beam at the anti-symmetric port is

$$I_{\text{AS}} = \Psi_{\text{AS}}^* \Psi_{\text{AS}}, \quad (1.5)$$

$$\simeq I_0 \phi_0^2 + 2I_0 \phi_0 \Delta\phi, \quad (1.6)$$

$$= I_0 \phi_0^2 + 2I_0 \phi_0 \frac{4\pi}{\lambda_{\text{EM}}} Lh, \quad (1.7)$$

In the absence of a gravitational wave signal, the DC light level at the antisymmetric port is related to the static offset by $\phi_0 = \sqrt{\frac{I_{\text{AS}}}{I_0}}$. As we are interested in a fluctuating signal, the DC offset from equation 1.7 can be removed, allowing for the signal at the antisymmetric port to be written as

$$I_{\text{AS}} = 2\sqrt{I_0 I_{\text{AS}}} \frac{4\pi}{\lambda_{\text{EM}}} Lh \quad (1.8)$$

and so by placing a photo detector in this beam, it is possible to measure the differential motion caused by the strain generated by the gravitational wave.

1.3.1 Using a Fabry Perot Cavity in the Arms of a Michelson Interferometer to enhance its Sensitivity

The optimal arm length of the interferometer is determined by 1/4 of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. For BBH and BNS, the gravitational wave signal will be at 100 Hz, and so the wavelength of the gravitational wave will be of the order 3000 km. Its not possible to construct such an interferometer on Earth.

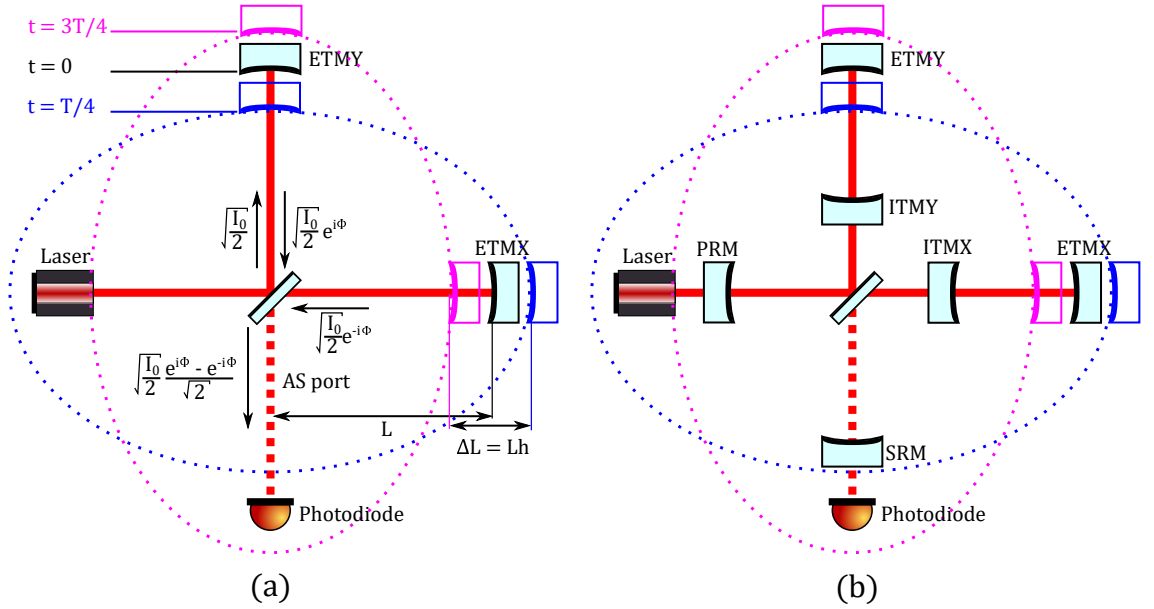


Figure 1.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different points during the gravitational wave's period is shown by the blue and pink lines. The difference in phase shift between the fields in the two arms affects the intensity of the light at the AS port. By detecting this light with a photodiode, the gravitational wave strain h can be determined. (b) To reach the sensitivity needed to detect gravitational wave, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

The sites that were available to construct the current generation of gravitational wave detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off the test mass, B , needs to be increased. Numerically, B should be ~ 250 . This means that equation 1.2 becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}BLh. \quad (1.9)$$

Fabry-Perot cavities are used² to increase the number of bounces of the light from the test masses. The cavity is formed by placing an *initial test mass* (ITM) near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. When a Fabry-Perot cavity goes through a resonance, the phase of the reflected light rapidly changes in response to a change in arm length. This is illustrated in Figure 1.2. The interferometer measures $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry Perot cavity is near or on resonance, as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry Perot than it is for a mirror, the signal is enhanced. The exact way this happens is dependent of the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity cavity as a low pass filter (e.g see [18, Chapter 3] or [19, Appendix D]).

²Herriott delay lines were considered as a means of achieving more bounces from the test masses, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry Perot cavities would perform equally well due to the Mizuno limit [18].

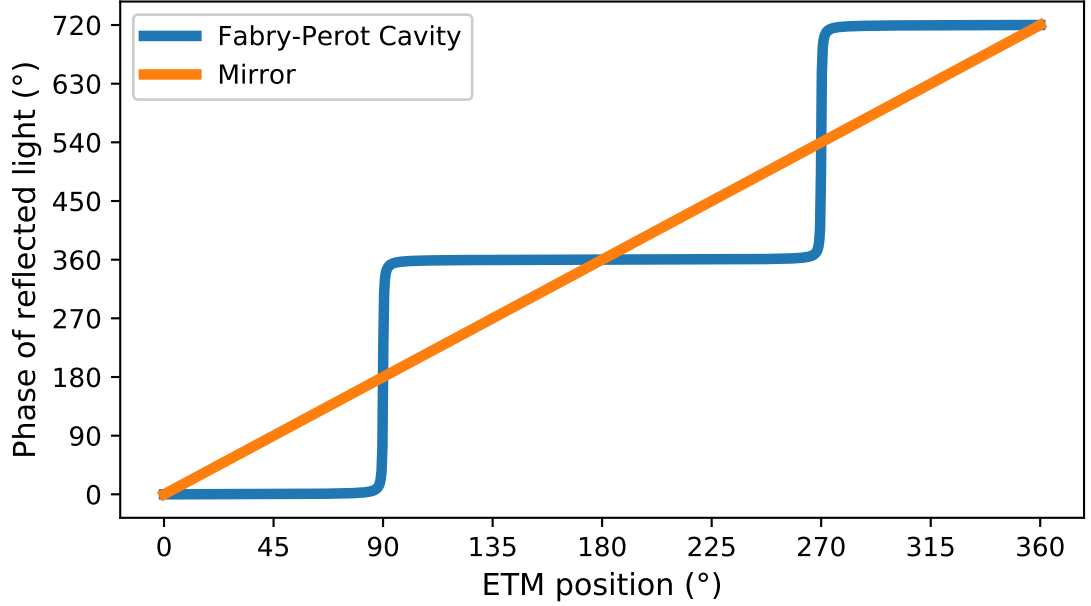


Figure 1.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity. The blue line shows the phase of the light that is reflected by a Fabry Perot cavity. Initially, the cavity is on anti-resonance so perfect destructive interference occurs as the round-trip phase is 180° . The end mirror (ETM) is moved 90° , resulting in an extra 180° round-trip phase, and so the cavity resonates. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The ETM is moved further, the cavity stops being resonant and the phase of the light is not changed as its reflected by the cavity. The orange line shows the phase of the light when a mirror is used instead of a cavity. The interferometers sensitivity is proportional to the slop $\frac{d\phi_{EM}}{d\phi_{ETM}}$, and so by inserting another mirror in the arm near the beamsplitter, a Fabry-Perot cavity is formed and the sensitivity of the interferometer is increased.

1.4 Sources of noise in Gravitational wave Interferometers.

As the gravitational wave strain causes a minuscule change in the phase accumulated by light in the arm cavities, the challenge of detecting gravitational waves lies in suppressing the noise below this signal. From equation 1.9, for a gravitational wave of strain $h_{\text{rms}} = 10^{-23}$ interacting with LIGO the change in phase shift that the light experiences is $\sim 10^{-11}$ rad.

1.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the detector. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The required energy to measure the phase shift that the light experiences due to the gravitational wave strain is determined by the time-energy uncertainty principle $\Delta\phi = \omega_{EM}\Delta T$ [20]. The measurement time, T , required corresponds to half the period of the gravitational wave $T = \frac{1}{2f_{\text{GW}}}$. The power of the laser is $I_0 = \hbar\omega_{EM}N$, where N is the number of photons detected in the time interval T , and so the energy used to measure the time is $E = I_0T = I_0\frac{\hbar\omega_{EM}N}{2f}$. As the laser is in a coherent state, the uncertainty in the number of photons detected in time T is $\Delta N = \sqrt{N}$. The uncertainty principle may now be written as

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (1.10)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the antisymmetric port will depend the

amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2I_{\text{AS}}\hbar\omega_{EM}\Delta f}. \quad (1.11)$$

The conversion from light power at the antisymmetric port to gravitational wave strain can be inferred from Equation 1.8

$$\frac{h}{I_{\text{AS}}} = \frac{1}{2\sqrt{I_0 I_{\text{AS}}}} \frac{\lambda_{\text{EM}}}{4\pi LB} \quad (1.12)$$

and so the amplitude spectrum of the strain corresponding to the the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c^2}{8I_0\omega_{EM}L^2B^2}\Delta f}. \quad (1.13)$$

Equation 1.13 assumed the frequency response to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector's noise (Figure 1.3). Equation 1.13 is more rigorously found in [21].

To beat the photon shot noise by a factor of 10, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave with a frequency of 100 Hz, the laser power required entering the arm cavity would need to be $\sim 1\text{MW}$. No such laser exists.

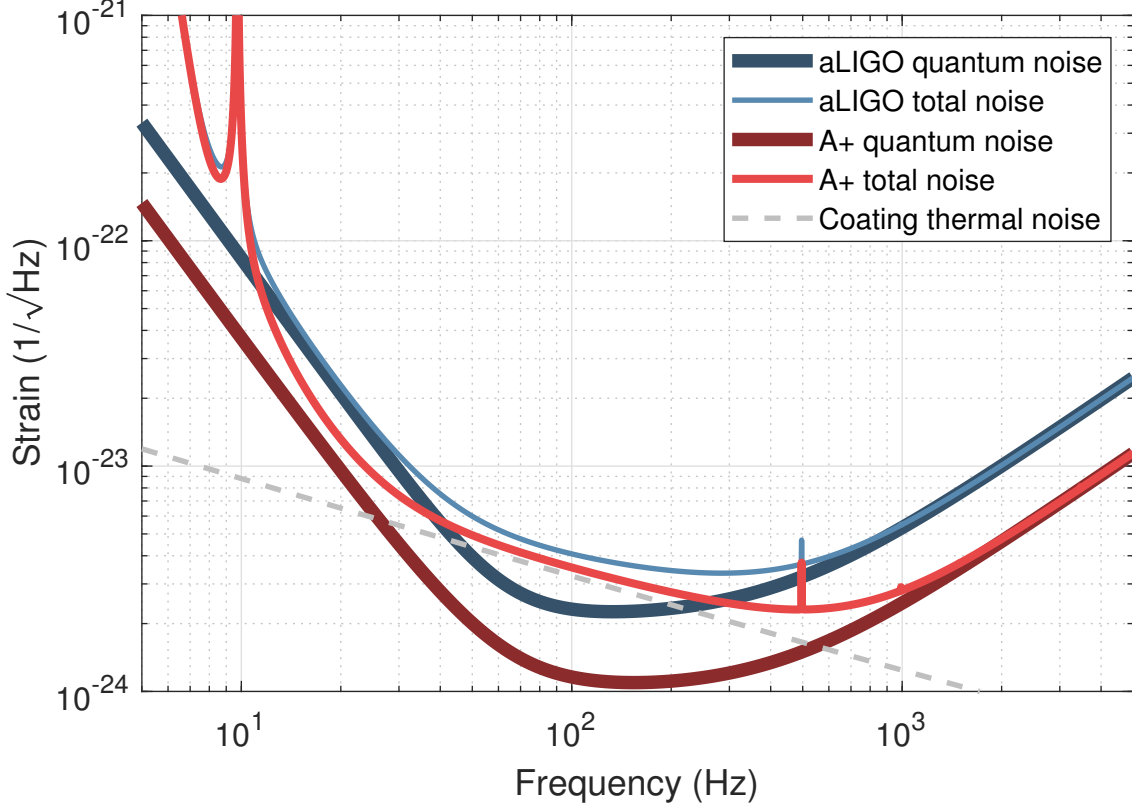


Figure 1.3: The strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, seismic noise begins to dominate the noise of the detector. Above 100 Hz, shot noise dominates the noise of the detector. This rises with f as the shot noise is filtered by the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure.

1.4.2 Power Recycling to reduce Shot Noise

To increase the laser power that is injected into the beam splitter, a power recycling cavity can be used [?]. Most of the light from the laser that is incident on the beam splitter gets reflected, effectively making the interferometer a mirror. Drever [?] proposed that a mirror, called the *power recycling mirror* (PRM), could be placed in the between the laser and the beam splitter and this would form a cavity. By tuning the position of the PRM such that the the light reflected from the interferometer and the light from the laser add in phase, the power would be ‘recycled’ and the total power impinging on the beam splitter would be increased. Current laser sources that can be operated for year-long time scales are capable of producing of the order 100W, and so the power recycling factor needs to be $\sim 10^4$. This technique has been successfully implemented at LIGO and by many other large scale interferometers. [?]

1.4.3 Signal recycling to shape quantum noise

The bandwidth of the detector depends on how long the signal carrying light is in the arms for. This detection bandwidth can be altered by introducing another mirror: the signal recycling mirror (SRM). This mirror is placed between the photodetector and the beamsplitter, and will form a coupled cavity with the arm cavities. The finesse of the arm cavities as seen by the light emerging at the antisymmetric port depends on the tuning of the SRM. There are two common modes of operation: resonant sideband extraction (RSE) [22, 23] and signal recycling .

In RSE mode, the signal recycling mirror is tuned so that the SRC is anti-resonant, thus broadening the sensitivity bucket at the expense of losing peak sensitivity.

This was motivated by wanting to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities (see Equation 1.13) . However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To increase the recover this lost bandwidth, the SRM is tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that SRC is resonant, deepening the sensitivity bucket at the expense of bandwidth. This comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth. This is known as the Mizuno limit [18].

1.4.4 Mode cleaning cavity to Remove Junk Light and Reduce Shot noise.

All light that is incident on the output photodiode contributes to the total noise of the measurement; therefore, only signal carrying light should be incident on the photodiode to enhance the signal to noise ratio of the measurement. In an ideal interferometer, the light emerging at the AS port carries only signal containing light and this beam will be in the *fundamental Gaussian mode*. However, due to optical imperfections and RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*.

To remove junk light, mode cleaners are used. A mode cleaner is a high finesse cavity that strips away higher order mode light and its control sidebands. Higher

order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity's finesse. The control sidebands are attenuated if they are outwith the cavities linewidth. The beam transmitted by the mode cleaner will be predominantly in the fundamental mode, and so the signal to noise ratio of the light measured by the photodiode will increase.

1.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test mass when it is reflected from it. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the ETMs' positions. The effect of radiation pressure noise on the LIGO detectors' sensitivity curves can be seen in Figure 1.3.

Radiation pressure noise scales with the quantum fluctuations entering the anti-symmetric port whereas shot noise is inversely proportional to these fluctuations. For a laser which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [24–26] sets an optimal the laser power for a given ETM mass which minimises the total quantum noise. The SQL forms a $1/f$ spectrum which; and for a certain laser power, the sensitivity curve a Michelson interferometer can only reach the SQL at only one frequency.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [?].

1.4.6 Coating Thermal Noise and other Noise Sources

There are many classical sources of noise to contend with in an gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of particles in the mirror coatings. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in earth crust, residual gas pressure noise that alters the refractive index of the arms, noise associated with the interferometer’s electronics and other technical noises.

1.5 Reducing Shot noise with Non-classical states of Light

The two-photon formalism developed by Caves and Schumaker [27,28] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO). A SHO can be described by two orthogonal components such as the momentum and position quadrature; for light a convenient basis to use is the *amplitude* and

phase quadrature. From this formalism, an uncertainty relation between phase and amplitude can be derived. If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signal we are trying to measure is in the phase quadrature of the light, it would be advantageous for, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature; this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting higher energy one photon into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at LLO [29] and GEO600 (citations).

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the antisymmetric port. During LIGO's third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise limiting the BNS range of the detector [29].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the intensity of the combined beam. The relative phase between

the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter ?? explores this in more detail.

1.6 Interferometry Techniques to beat Quantum Limits

Squeezed light and the natural correlations, called ponderomotive squeezing [25], created in the interferometer can be used to surpass the SQL at some frequencies. This has been demonstrated the LIGO Livingston detector [30]. However, this technique results in noise at other frequencies being enhanced.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency-dependent squeezer can be used. The light fields squeezing angle can be rotated as a function of frequency. This is done by using a detuned filter cavity. The light injected into it is rotated depending on its frequency. At low frequencies, the amplitude quadrature can be squeezed and the radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and the shot noise can be decreased. This will be part of the A+ detector.

To beat this bandwidth limit, energy has to be injected into the system. White light cavities as a means of increasing the detector bandwidth have been studied (cite Haixing), although these solutions have yet to be demonstrated in the lab. Alternatively, PT-symmetry schemes may be used to beat the bandwidth limit, again these have yet to be demonstrated in the lab.

There are several sub-SQL schemes. To avoid radiation pressure noise, a momentum measurement of the test masses could be made. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$. A random impartation of ‘position’ would not affect future measurements. Such measurements are known as quantum non-demolition. [variational readout](#)

1.7 The Effect Losses have on Measuring Quantum States of Light

To preserve the benefits gained by using quantum states of light, it is essential to have minimal losses in the read-out of a gravitational wave detector. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss. The incoming enters side of the beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal component, and so the state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss [31].

Therefore, it is critical that the photodiode used to measure the AS beam has low loss. Photodiodes convert photons into electrons, ideally with a one-to-one ratio. The measure of how well this is done is the photodiode’s *quantum efficiency*, η . The factor by which the shot noise increases, η , due to a simple loss mechanism, such

as a non-ideal photodiode, is

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}. \quad (1.14)$$

It is acceptable in a gravitational wave interferometer to have a quantum efficiency of at least 99%. This can be achieved by photodiodes fabricated from InGaAs.

Mode matching losses increase have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism (cite lee's work). This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss.

1.8 Overview of the LIGO detector

The LIGO detector is a dual-recycled-Fabry-Perot Michelson interferometer with 4km arms. The layout of the detector is shown in Figure 1.4 and the noise spectral density is shown in Figure 1.3. The term dual-recycled refers to the fact that the interferometer uses both power and, the mathematical dual of power recycling, signal recycling. The interferometer is housed within a vacuum system comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 1.4 includes all the optics from the pre-stabilised-laser (PSL) to the readout photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors

in addition to the PRM known as PR2 and PR3. The role of these mirrors is to make round trip gowy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason. The beam passes through the output Faraday isolator (OFI), and is steered into the output mode cleaner. The gravitational signal is encoded on the beam transmitted by the OMC, so photodiode monitor the light transmitted by the OMC.

1.9 The future of Ground based Gravitational Wave Detectors

Slight repetition with section 1.2 The next generation of ground based gravitational wave detectors include the Einstein telescope and LIGO cosmic explorer. These interferometers will be ten times larger than the current generation of interferometers and will be able to test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic signals which have not yet been theorised.

As the limiting sources of of current gravitational wave detectors is the coating thermal noise, research into cryogenic mirror coatings is underway. Fused silica is opaque at these frequencies and does the noise characteristics do not improve when cooled, so silicon test masses are being developed.

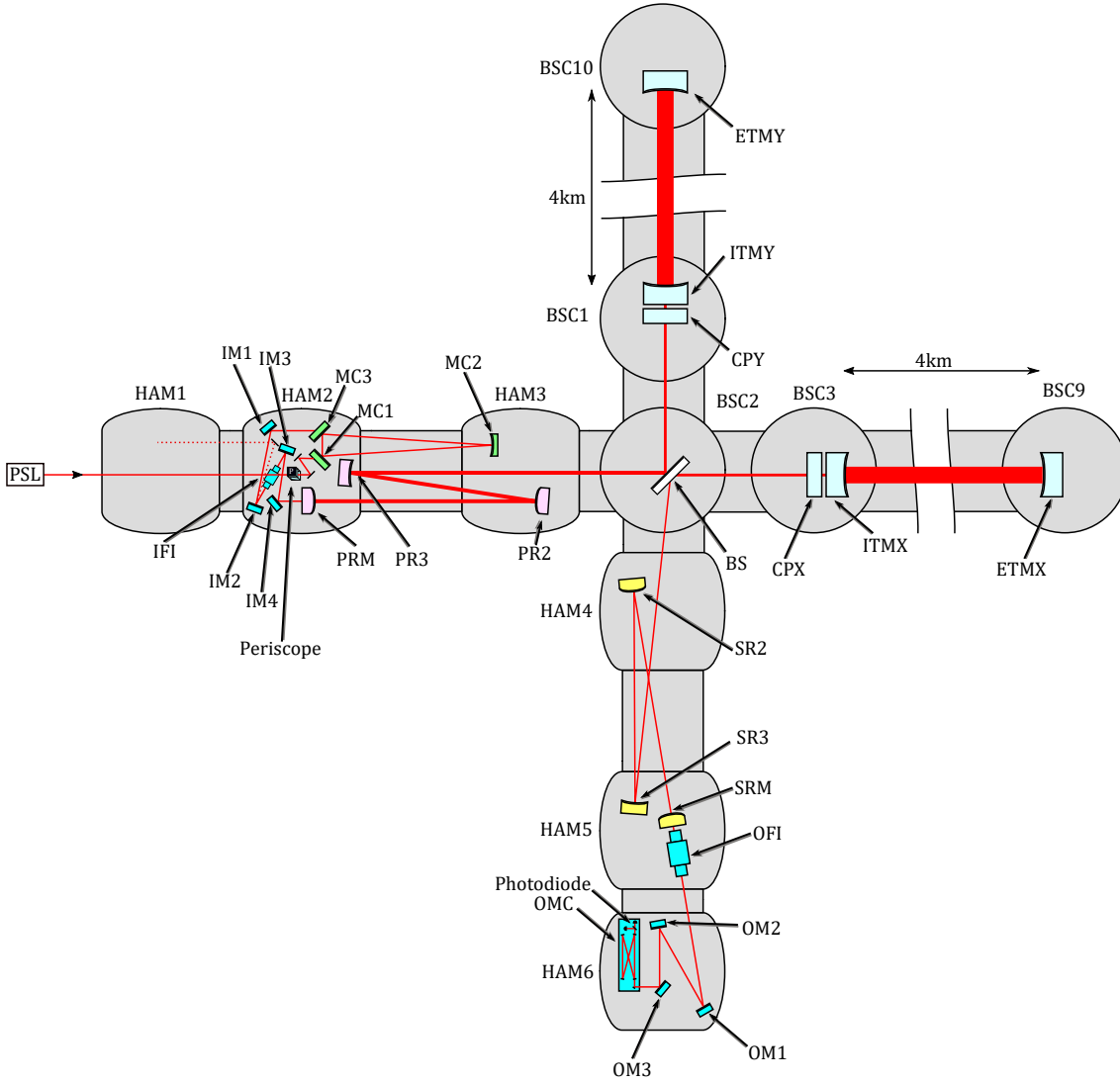


Figure 1.4: Sketch of a LIGO detector. The input and output optics, IM1-4, input periscope, OM1-3, OMC, IFI and OFI, are shown in blue. The input mode cleaner is shown in green. The power recycling is shown in pink. The arm cavities are shown in pale blue. The signal recycling cavity is shown in yellow.

It is expected that low noise, low loss coatings will not be compatible with 1 μm light, so these detectors will operate with either a 1.5 μm or 2 μm wavelength laser. Laser technology for 2 μm light is still in its infancy, and so there is lots of uncertainty in the practicality of using 2 μm as the operating wavelength of a gravitational wave detector; for instance, low noise, high quantum efficiency photodiodes that can detect 2 μm light have never been fabricated.

1.10 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. In order for this to be successfully implemented, the quantum state of the light field must be accurately detected with a balanced homodyne readout and losses in the read-out must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with 2 μm light, future detectors may use a 2 μm laser; this leads to the question: is laser technology at this wavelength sufficiently mature enough to enable quantum limited measurements? Balanced homodyne detection and photodiodes with high quantum efficiency and low dark noise will be required for any future interferometer.

Bibliography

- [1] LIGO Virgo Collaboration. Observation of gravitational waves from a binary black hole merger. *Phys. Rev. Lett.*, 116:061102, Feb 2016.
- [2] LIGO Scientific and Virgo Collaboration. The basic physics of the binary black hole merger GW150914. *Annalen der Physik*, 529(1-2):1600209, oct 2016.
- [3] LIGO Virgo Collaboration. Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Phys. Rev. Lett.*, 119:161101, Oct 2017.
- [4] The LVK collaboration. GW190814: Gravitational waves from the coalescence of a 23 solar mass black hole with a 2.6 solar mass compact object. *The Astrophysical Journal*, 896(2):L44, jun 2020.
- [5] The LIGO Virgo Collaboration. Gwtc-2: Compact binary coalescences observed by ligo and virgo during the first half of the third observing run.
- [6] The LIGO Scientific Collaboration and the Virgo Collaboration. Tests of general relativity with binary black holes from the second ligo-virgo gravitational-wave transient catalog.
- [7] Charles W. Misner. *Gravitation*. Princeton University Press, oct 2017.

- [8] LVK collaboration. Narrow-band search for gravitational waves from known pulsars using the second LIGO observing run. *Physical Review D*, 99(12), jun 2019.
- [9] The LIGO Virgo Collaboration. Searches for continuous gravitational waves from 15 supernova remnants and fomalhaut b with advanced LIGO. *The Astrophysical Journal*, 875(2):122, apr 2019.
- [10] The LIGO Virgo Collaboration. Search for transient gravitational-wave signals associated with magnetar bursts during advanced LIGO’s second observing run. *The Astrophysical Journal*, 874(2):163, apr 2019.
- [11] Ernazar Abdikamalov, Giulia Pagliaroli, and David Radice. Gravitational waves from core-collapse supernovae.
- [12] T. Regimbau, M. Evans, N. Christensen, E. Katsavounidis, B. Sathyaprakash, and S. Vitale. Digging deeper: Observing primordial gravitational waves below the binary-black-hole-produced stochastic background. *Physical Review Letters*, 118(15), apr 2017.
- [13] The LIGO Virgo Collaboration. Search for the isotropic stochastic background using data from advanced LIGO’s second observing run. *Physical Review D*, 100(6), sep 2019.
- [14] Juan García-Bellido. Massive primordial black holes as dark matter and their detection with gravitational waves. *Journal of Physics: Conference Series*, 840:012032, may 2017.

- [15] Simeon Bird, Ilias Cholis, Julian B. Muñoz, Yacine Ali-Haïmoud, Marc Kamionkowski, Ely D. Kovetz, Alvise Raccanelli, and Adam G. Riess. Did LIGO detect dark matter? *Physical Review Letters*, 116(20), may 2016.
- [16] Yanou Cui, Marek Lewicki, David E. Morrissey, and James D. Wells. Probing the pre-BBN universe with gravitational waves from cosmic strings. *Journal of High Energy Physics*, 2019(1), jan 2019.
- [17] Gregory M Harry, Matthew R Abernathy, Andres E Becerra-Toledo, Helena Armandula, Eric Black, Kate Dooley, Matt Eichenfield, Chinyere Nwabugwu, Akira Villar, D R M Crooks, Gianpietro Cagnoli, Jim Hough, Colin R How, Ian MacLaren, Peter Murray, Stuart Reid, Sheila Rowan, Peter H Sneddon, Martin M Fejer, Roger Route, Steven D Penn, Patrick Ganau, Jean-Marie Mackowski, Christophe Michel, Laurent Pinard, and Alban Remillieux. Titania-doped tantala/silica coatings for gravitational-wave detection. *Classical and Quantum Gravity*, 24(2):405–415, dec 2006.
- [18] J. Mizuno. *Comparison of optical configurations for laser-interferometric gravitational-wave detectors*. PhD thesis, AEI-Hannover, MPI for Gravitational Physics, Max Planck Society, 1995.
- [19] Gerhard Heinzl. Advanced optical techniques for laser-interferometric gravitational-wave detectors, 1999.
- [20] Vladimir B. Braginsky. Energetic quantum limit in large-scale interferometers. In *AIP Conference Proceedings*. AIP, 2000.
- [21] B.J. Meers. The frequency response of interferometric gravitational wave detectors. *Physics Letters A*, 142(8-9):465–470, dec 1989.

- [22] G. Heinzel, J. Mizuno, R. Schilling, W. Winkler, A. Rüdiger, and K. Danzmann. An experimental demonstration of resonant sideband extraction for laser-interferometric gravitational wave detectors. *Physics Letters A*, 217(6):305–314, jul 1996.
- [23] J. Mizuno, K.A. Strain, P.G. Nelson, J.M. Chen, R. Schilling, A. Rüdiger, W. Winkler, and K. Danzmann. Resonant sideband extraction: a new configuration for interferometric gravitational wave detectors. *Physics Letters A*, 175(5):273–276, apr 1993.
- [24] Carlton M. Caves. Quantum-mechanical noise in an interferometer. *Physical Review D*, 23(8):1693–1708, apr 1981.
- [25] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [26] V. B. Braginsky and F. Ya. Khalili. Quantum nondemolition measurements: the route from toys to tools. *Reviews of Modern Physics*, 68(1):1–11, jan 1996.
- [27] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.
- [28] Bonny L. Schumaker and Carlton M. Caves. New formalism for two-photon quantum optics. II. mathematical foundation and compact notation. *Physical Review A*, 31(5):3093–3111, may 1985.

- [29] M. Tse, Haocun Yu, N. Kijbunchoo, A. Fernandez-Galiana, P. Dupej, L. Barsotti, C.D. Blair, D.D. Brown, S.E. Dwyer, A. Effler, M. Evans, P. Fritschel, V.V. Frolov, A.C. Green, G.L. Mansell, F. Matichard, N. Mavalvala, D.E. McClelland, L. McCuller, T. McRae, J. Miller, A. Mullavey, E. Oelker, I.Y. Phinney, D. Sigg, B.J.J. Slagmolen, T. Vo, R.L. Ward, C. Whittle, R. Abbott, C. Adams, R.X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J.S. Areeda, Y. Asali, S.M. Aston, C. Austin, A.M. Baer, M. Ball, S.W. Ballmer, S. Bannagiri, D. Barker, J. Bartlett, B.K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R.M. Blair, N. Bode, P. Booker, R. Bork, A. Bramley, A.F. Brooks, A. Buikema, C. Cahillane, K.C. Cannon, X. Chen, A.A. Ciobanu, F. Clara, S.J. Cooper, K.R. Corley, S.T. Countryman, P.B. Covas, D.C. Coyne, L.E.H. Datrier, D. Davis, C. Di Fronzo, J.C. Driggers, T. Etzel, T.M. Evans, J. Feicht, P. Fulda, M. Fyffe, J.A. Giaime, K.D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray, Anchal Gupta, E.K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hardwick, R.K. Hasskew, M.C. Heintze, A.F. Helmling-Cornell, N.A. Holland, J.D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, P.J. King, J.S. Kissel, Rahul Kumar, M. Landry, B.B. Lane, B. Lantz, M. Laxen, Y.K. Lecoecueche, J. Leviton, J. Liu, M. Lormand, A.P. Lundgren, R. Macas, M. MacInnis, D.M. Macleod, S. Márka, Z. Márka, D.V. Martynov, K. Mason, T.J. Massinger, R. McCarthy, S. McCormick, J. McIver, G. Mendell, K. Merfeld, E.L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno, C.M. Mow-Lowry, S. Mozzon, T.J.N. Nelson, P. Nguyen, L.K. Nuttall, J. Oberling, R.J. Oram, B. O'Reilly, C. Osthelder, D.J. Ottaway, H. Overmier, J.R. Palamos, W. Parker, E. Payne, A. Pele, C.J. Perez, M. Pirello, H. Radkins, K.E. Ramirez, J.W. Richard-

son, K. Riles, N. A. Robertson, J. G. Rollins, C. L. Romel, J. H. Romie, M. P. Ross, K. Ryan, T. Sadecki, E. J. Sanchez, L. E. Sanchez, T. R. Saravanan, R. L. Savage, D. Schaetzel, R. Schnabel, R. M. S. Schofield, E. Schwartz, D. Sellers, T. J. Shaffer, J. R. Smith, S. Soni, B. Sorazu, A. P. Spencer, K. A. Strain, L. Sun, M. J. Szczepańczyk, M. Thomas, P. Thomas, K. A. Thorne, K. Toland, C. I. Torrie, G. Traylor, A. L. Urban, G. Vajente, G. Valdes, D. C. Vanderhyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, C. Vorvick, M. Wade, J. Warner, B. Weaver, R. Weiss, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, M. J. Yap, Hang Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Quantum-enhanced advanced LIGO detectors in the era of gravitational-wave astronomy. *Physical Review Letters*, 123(23), dec 2019.

- [30] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [31] Stefan L. Danilishin and Farid Ya. Khalili. Quantum measurement theory in gravitational-wave detectors. *Living Reviews in Relativity*, 15(1), apr 2012.